

EE326 Intro. to Information Theory and Coding Final

Spring 2024

Date: Monday, June 10, 2024

Time: 9am-11:30am

Problem 1 (10 points)

In this problem, we consider some variants of the discrete memoryless channel.

- a) (5 points) Let's consider a channel with memory. Consider a binary symmetric channel with $Y_i = X_i \oplus Z_i$, where $\oplus$ is mod 2 addition, and $X_i, Y_i \in \{0, 1\}$. Suppose that $\{Z_i\}$ has constant marginal probabilities $\Pr(Z_i = 1) = p = 1 - \Pr(Z_i = 0)$, but that $Z_1, Z_2, \dots, Z_n$ are not necessarily independent. Assume that Z^n is independent of the input X^n . Let $C = 1 - H_B(p)$. Compare the size of $\max_{P(X_1, X_2, \dots, X_n)} I(X_1, X_2, \dots, X_n; Y_1, Y_2, \dots, Y_n)$ and nC . Which one is bigger?

$$\max_{P(X_1, X_2, \dots, X_n)} I(X_1, X_2, \dots, X_n; Y_1, Y_2, \dots, Y_n) \geq nC \text{ or,}$$

$$\max_{P(X_1, X_2, \dots, X_n)} I(X_1, X_2, \dots, X_n; Y_1, Y_2, \dots, Y_n) \leq nC ?$$

Choose your answer and prove your statement.

- b) (5 points) For this problem, let's consider a channel with two independent observations of output Y . Let Y_1 and Y_2 be conditionally independent and identically distributed given input X . Compare the size of $I(X; Y_1, Y_2)$ and $2I(X; Y_1)$. Which one is bigger?

$$I(X; Y_1, Y_2) \geq 2I(X; Y_1) \text{ or,}$$

$$I(X; Y_1, Y_2) \leq 2I(X; Y_1) ?$$

Choose your answer and prove your statement.

Problem 2 (15 points)

Consider the channel with $x, y \in \{0, 1, 2, 3\}$ and transition probabilities $P_{Y|X}(y|x)$ given by the following matrix:

$$\begin{bmatrix} P_{Y|X}(0,0) & P_{Y|X}(0,1) & P_{Y|X}(0,2) & P_{Y|X}(0,3) \\ \vdots & \vdots & \vdots & \vdots \\ P_{Y|X}(3,0) & P_{Y|X}(3,1) & P_{Y|X}(3,2) & P_{Y|X}(3,3) \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 1/2 & 0 & 0 & 1/2 \end{bmatrix}$$

- a) (3 points) Find the capacity of this channel.
- b) (4 points) Define the random variable $Z = g(Y)$ where

$$g(Y) = \begin{cases} A & \text{if } Y \in \{0, 1\} \\ B & \text{if } Y \in \{2, 3\}. \end{cases}$$

For the following two distributions for X , compute $I(X; Z)$:

(i)

$$P_X(x) = \begin{cases} 1/2 & \text{if } x \in \{1, 3\} \\ 0 & \text{if } x \in \{0, 2\}. \end{cases}$$

(ii)

$$P_X(x) = \begin{cases} 0 & \text{if } x \in \{1, 3\} \\ 1/2 & \text{if } x \in \{0, 2\}. \end{cases}$$

- c) (4 points) Find the capacity of the channel between X and Z , specifically where $X \in \{0, 1, 2, 3\}$ and $Z \in \{A, B\}$, and the transition probabilities $P_{Z|X}(z|x)$ are given by

$$P_{Z|X}(z|x) = \sum_{y_0: g(y_0)=z} P_{Y|X}(y_0|x).$$

- d) (4 points) For the X distribution of part (ii) of b), does $X \rightarrow Z \rightarrow Y$ form a Markov chain? Given reasoning to your answer.

Problem 3 (15 points)

In class, we discussed the design of Generative Adversarial Networks (GAN) with the optimization:

$$\min_{G(\cdot)} \max_{D(\cdot)} \sum_{z \in \mathcal{Y} \cup \hat{\mathcal{Y}}} \mathbb{Q}_Y(z) \log D(z) + \mathbb{Q}_{\hat{Y}}(z) \log(1 - D(z))$$

where $\mathbb{Q}_Y(\cdot)$ and $\mathbb{Q}_{\hat{Y}}(\cdot)$ denote the distributions for real data and fake (generated) data, respectively. We also showed the optimal discriminator $D^*(\cdot)$ maximizing the above inner optimization is

$$D^*(z) = \frac{\mathbb{Q}_Y(z)}{\mathbb{Q}_Y(z) + \mathbb{Q}_{\hat{Y}}(z)}.$$

Plugging in this discriminator function, the optimization for the generator $G(\cdot)$ becomes

$$G_{\text{GAN}}^* = \arg \min_{G(\cdot)} \frac{1}{2} (D_{\text{KL}}(\mathbb{Q}_Y \| (\mathbb{Q}_Y + \mathbb{Q}_{\hat{Y}})/2) + D_{\text{KL}}(\mathbb{Q}_{\hat{Y}} \| (\mathbb{Q}_Y + \mathbb{Q}_{\hat{Y}})/2)).$$

a) (5 points) The quantity

$$D_{JS}(\mathbb{Q}_Y \| \mathbb{Q}_{\hat{Y}}) := \frac{1}{2} (D_{\text{KL}}(\mathbb{Q}_Y \| (\mathbb{Q}_Y + \mathbb{Q}_{\hat{Y}})/2) + D_{\text{KL}}(\mathbb{Q}_{\hat{Y}} \| (\mathbb{Q}_Y + \mathbb{Q}_{\hat{Y}})/2))$$

is called Jensen-Shannon divergence. Show the equivalence between Jensen-Shannon divergence and the following mutual information

$$I(X; V) = D_{JS}(\mathbb{Q}_Y \| \mathbb{Q}_{\hat{Y}})$$

where V is a random variable uniformly distributed over $\{0, 1\}$, and conditioned on $V = 0$, X is drawn from the distribution $\mathbb{Q}_Y$ and conditioned on $V = 1$, X is drawn from the distribution $\mathbb{Q}_{\hat{Y}}$.

b) (5 points) Find two constants c_1, c_2 (the maximum possible c_1 and minimum possible c_2) with which we can always bound the Jensen-Shannon divergence:

$$c_1 \leq D_{JS}(\mathbb{Q}_Y \| \mathbb{Q}_{\hat{Y}}) \leq c_2.$$

Find the conditions for achieving the equality in the lower bound and the equality in the upper bound, respectively.

c) (5 points) Let's consider a scenario where your training of GAN was somewhat failed so that GAN generates samples from a distribution $\mathbb{Q}_{\hat{Y}}$ different from the true distribution $\mathbb{Q}_Y$, i.e., $\mathbb{Q}_{\hat{Y}} \neq \mathbb{Q}_Y$. But, assume that, fortunately, $M := \max_z \frac{\mathbb{Q}_Y(z)}{\mathbb{Q}_{\hat{Y}}(z)}$ is bounded and this value is known. Then, we will design a rejection sampling scheme such that whenever generating a new sample $z \sim \mathbb{Q}_{\hat{Y}}$, we will either reject the sample or accept the sample, depending on the following strategy: sample a value $u \sim \text{Unif}[0, 1]$ and if

$$u \begin{cases} \leq \frac{\mathbb{Q}_Y(z)}{M\mathbb{Q}_{\hat{Y}}(z)} & \text{accept the sample;} \\ \geq \frac{\mathbb{Q}_Y(z)}{M\mathbb{Q}_{\hat{Y}}(z)} & \text{reject the sample.} \end{cases} \quad (1)$$

The problem is that we do not know the distributions $\mathbb{Q}_Y(\cdot)$ and $\mathbb{Q}_{\hat{Y}}(\cdot)$. Come up with an idea to use the discriminator output $D^*(z)$ to implement this rejection sampling.

d) (Bonus 3 points) Show why the above rejection sampling can provide you samples that have the same distribution as the desired distribution $\mathbb{Q}_Y(\cdot)$.